Snowflake Project – Data Engineering & SQL Solution

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	https://github.com/shamailairfan317-cpu/Global-Olympic-Sports-Performance-Analysis
Data Source(s) Link	Historical Olympic Datasets (medals_results.csv, athlete.csv, athlete_summary.csv)
Database	OLYMPIC_PERFORMANCE_DB (Cloud Data Warehouse)
Warehouse	COMPUTE_WH (Size: X-Small / Standard Virtual Warehouse)
Tables / Views Created	dim_athletes, fact_medals, vw_country_medal_tally, vw_athlete_participation_trend, vw_gender_distribution
Key Transformations / Calculations	Cleaned missing values in age, height, and weight columns; merged athlete event records with medal tables; calculated cumulative gold/silver/bronze ratios and

	gender participation percentages using SQL & Pandas aggregations.
KPIs created	Total Gold/Silver/Bronze Medal Counts, All-Time Leaderboard Rankings, Athlete Turnout Growth Rate, Female-to-Male Participation Ratio, and Sport-wise Age Benchmarks.
Key Insights	Identified dominant sports powerhouses (USA, URS, GER), tracked the exponential growth of female athlete participation from early 20th century to modern games, and correlated physical attributes (height/weight clusters) with medal success in sports like Basketball and Gymnastics.

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	yes
2	Data is cleaned, transformed, and prepared appropriately	yes
3	Required tables, views, and data relationships are correctly implemented	yes
4	Required SQL logic, calculations, and KPIs are correctly implemented	yes
5	Data and solution outputs are validated against the project requirements	yes
6	Solution is properly organized, documented, and functional	yes

Additional Notes / Comments:

Pipeline Execution: Data was ingested from raw CSV files into staging tables, transformed using SQL scripts for deduplication and schema normalization, and queried to power interactive visual analytics dashboards.

Verification: All metric outputs and analytical queries were cross-verified against expected historical Olympic datasets to guarantee data integrity and accuracy prior to final submission.